

FIGURE P1-13

- 1-14 An incandescent lamp absorbs 100 W when connected to a 120 V source. A fluorescent lamp producing the same amount of light absorbs 12 W when connected to the same source. How much cheaper is it to operate the fluorescent bulb rather than the incandescent bulb over 1000 hours when electricity costs 7.2 cents/kWh?
- 1-15 The current through a device is zero for $t < 0$ and is $i(t) = 3e^{-2t}$ A for $t \geq 0$. Find the charge $q(t)$ flowing through the device for $t \geq 0$.
- 1-16 The current through and voltage across a two-terminal device are $i(t) = 20 \sin(1000t)$ mA and $v = 100 \sin(1000t)$ V.

FIGURE P1-21

1-22 Figure P1-22 shows an electric circuit with a voltage and current variable assigned to each of the six devices. The device voltages and currents are observed to be

	v (V)	i (A)
Device 1	15	-1
Device 2	5	1
Device 3	10	2
Device 4	-10	-1
Device 5	20	-3
Device 6	20	2

Find the power associated with each device and state whether the device is absorbing or delivering power. Use the power balance to check your work.

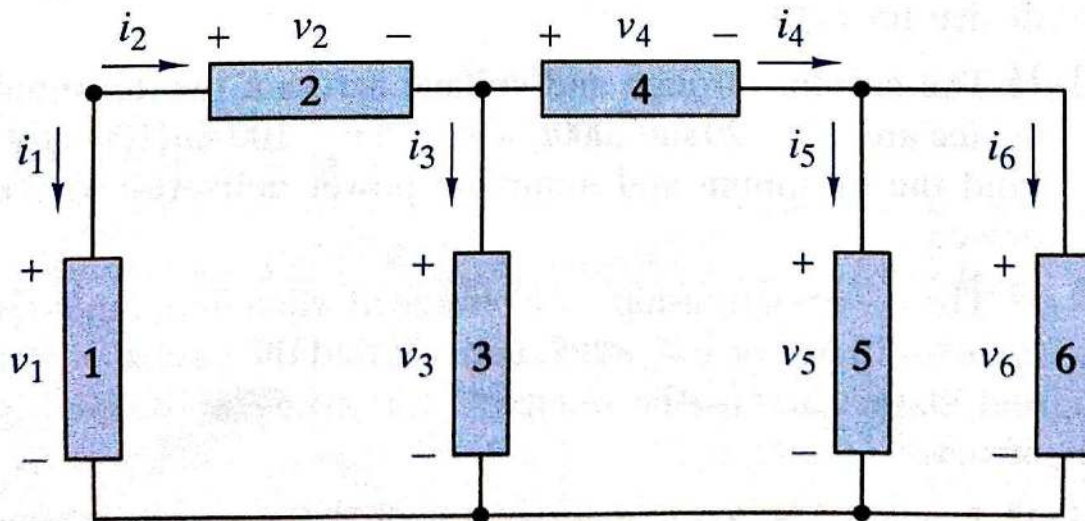


FIGURE P1-22

2–5 In Figure P2–5 find R_x and the power delivered to the resistor.

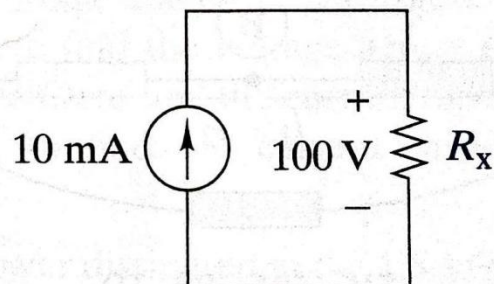


FIGURE P2–5

2–6 The i – v characteristic of a nonlinear resistor is $v = 75i + 0.2i^3$.

(a) Calculate v and p for $i = \pm 0.5, \pm 1, \pm 2, \pm 5$, and ± 10 A.

(b) Find the maximum error in v when the device is treated

2-13 For the circuit in Figure P2-13:

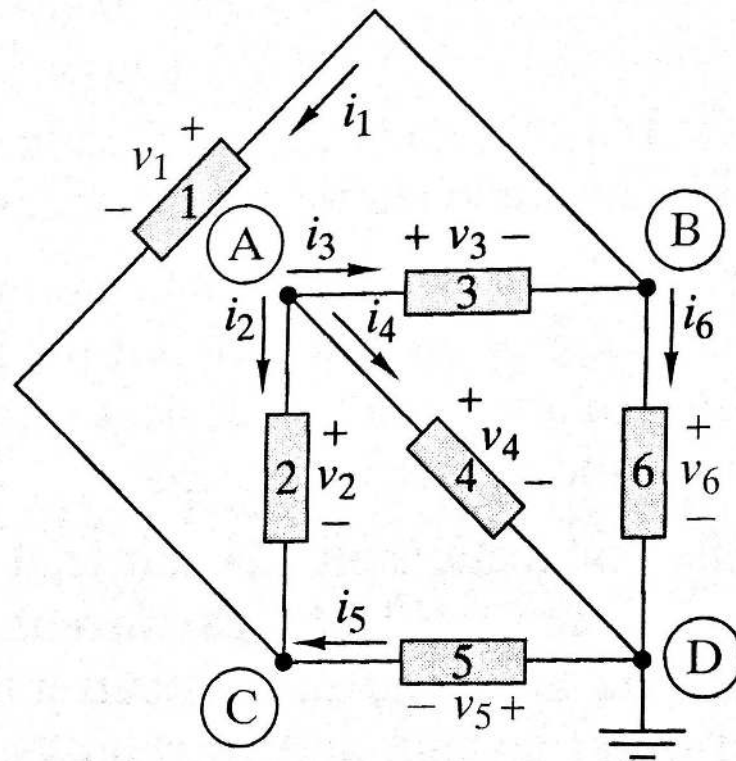


FIGURE P2-13

(a) Identify the nodes and at least three loops in the circuit.

(c) Write KCL and KVL connection equations for the circuit.

2-14 In Figure P2-13 $v_2 = 10\text{ V}$, $v_3 = -10\text{ V}$, and $v_4 = 3\text{ V}$. Find v_1 , v_5 , and v_6 .

2-15 The circuit in Figure P2-15 is organized around the three signal lines A, B, and C.

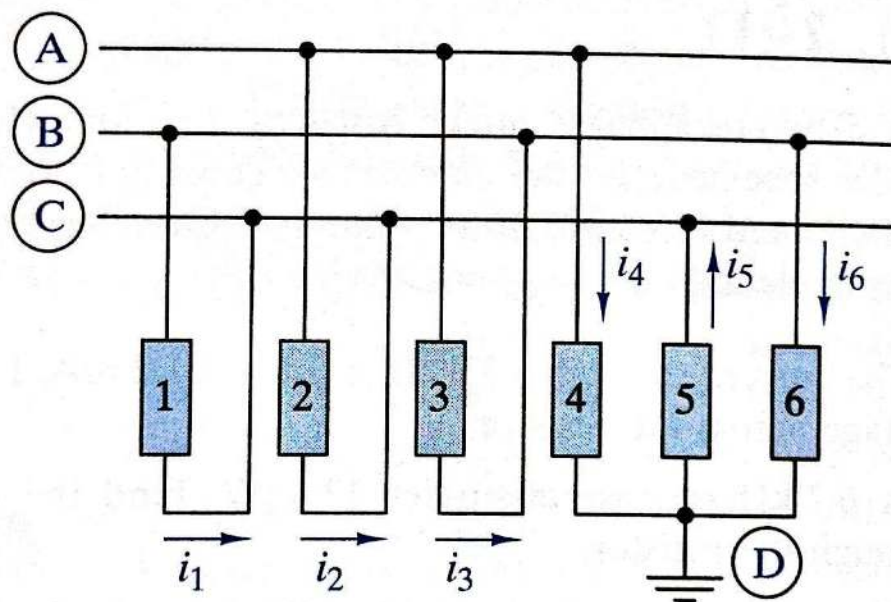


FIGURE P2-15

(a) Identify the nodes and at least three loops in the circuit.

(b) Write KCL connection equations for the circuit.

(c) If $i_1 = -20\text{ mA}$, $i_2 = -12\text{ mA}$, and $i_3 = 50\text{ mA}$, find i_4 , i_5 , and i_6 .

(d) Show that the circuit in Figure P2-15 is identical to that in Figure P2-13.

2-16 In Figure P2-16 $v_2 = 10\text{ V}$, $v_3 = 10\text{ V}$, and $v_4 = 10\text{ V}$. Find v_1 and v_5 .

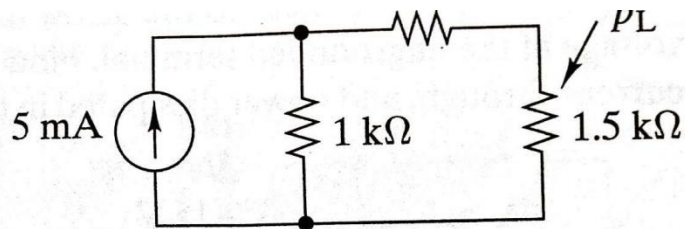


FIGURE P2-23

2-24 Figure P2-24 shows a subcircuit connected to the rest of the circuit at four points.

- (a) Use element and connection constraints to find v_x and i_x .
- (b) Show that the sum of the currents into the rest of the circuit is zero.

(c) Find the voltage v_A with respect to the ground in the circuit.

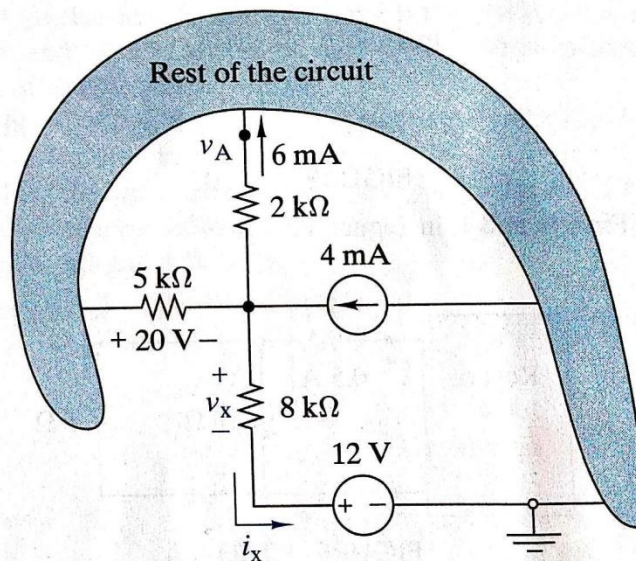


FIGURE P2-24

OBJECTIVE 2-4 (SECT. 2-4)

- (a) Given a circuit, find an equivalent resistance.
 - (b) Given a circuit, find an equivalent circuit.
- See Examples 2-11 through 2-14.

2-27 Find the equivalent resistance R_{eq} looking into the terminals a and b in the circuit in Figure P2-27.

R_{eq}

2-28 Find the equivalent resistance R_{eq} looking into the terminals a and b in the circuit in Figure P2-28.

R_{eq}

2-25 In Figure P2-25 $i_x = -0.5$ mA. Find the value of R .

FIGURE P2-36

5 k Ω .
2-37 Two 10-k Ω potentiometers (a potentiometer is a variable resistor whose value between the two ends is 10 k Ω and between one end and the wiper—the third terminal—can range from 0 Ω to 10 k Ω) are connected as shown in Figure P2-37. What is the range of R_{EQ} ?

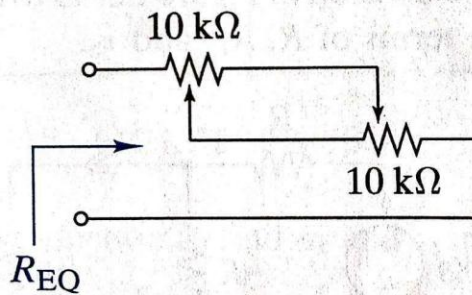
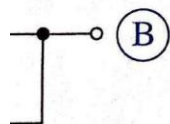


FIGURE P2-37

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2-38 Select the value of R in Figure P2-38 so that $R_{AB} = R_L$.



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Exercises 2–10,

btain an expres-

- 2–44 The $1\text{-k}\Omega$ load in Figure P2–44 needs 5 V across it to operate correctly. Where should the wiper on the potentiometer be set (R_X) to obtain the desired output voltage?

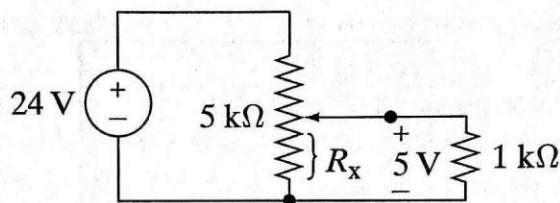


FIGURE P2–44

- 2–45 Find the range of values of v_O in Figure P2–45.

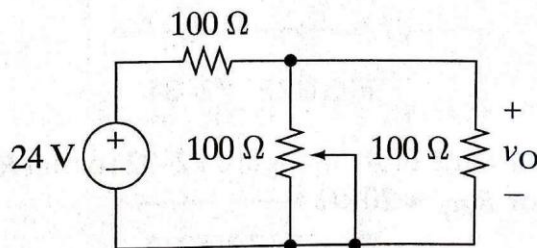


FIGURE P2–45

2–46 Figure P2–46 shows a voltage bridge circuit, that is, two

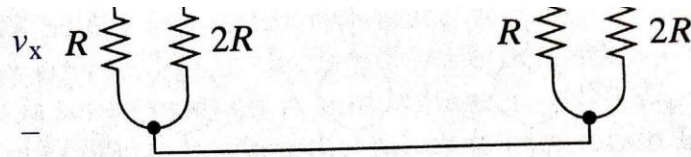


FIGURE P2-52

2-53 Use circuit reduction to find v_x and the power provided by the source in Figure P2-53.

$v_L = 3 \text{ V}$.

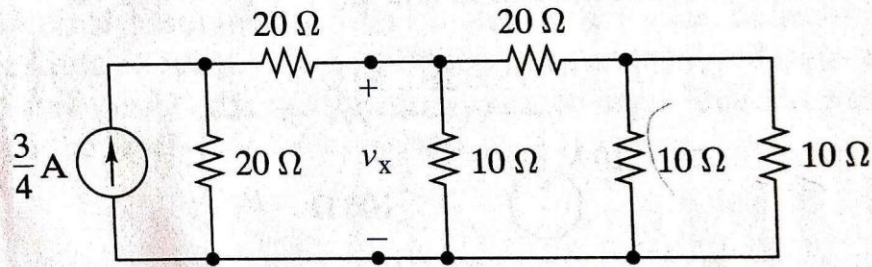


FIGURE P2-53

at $v_L = 2 \text{ V}$.
positive.

2-54 Use circuit reduction to find v_x and i_x in Figure P2-54.



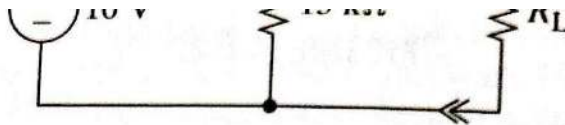


FIGURE P3-35

- 3-36(a)** Find the Thévenin or Norton equivalent circuit seen by R_L in Figure P3-36.
(b) Use the equivalent circuit found in part (a) to find i_L in terms of i_S , R_1 , R_2 , and R_L .
(c) Check your answer to part (b) using current division.

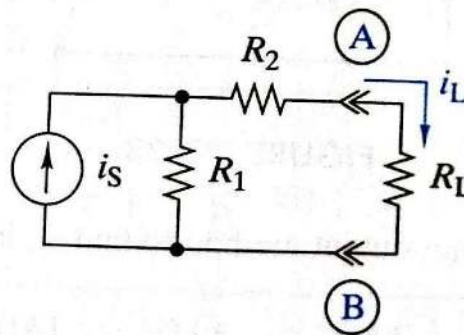


FIGURE P3-36

- 3-37** Find the Thévenin equivalent circuit seen by R_L in Figure P3-37. Find the voltage across the load when $R_L = 5 \Omega$, 10Ω , and 50Ω .

20 Ω

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when $R_L =$

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is equal to 37

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in both situa-

3-42 Assume that Figure P3-42 represents a model of the auxiliary output port of a car. The output current is $i = 1$ A when $v = 0$ V. The output voltage is $v = 12$ V when a $20\text{-}\Omega$ resistor is connected between the terminals. Suppose you wanted to charge a 12-V battery by connecting the battery at the port. How much current would the port deliver to the battery?

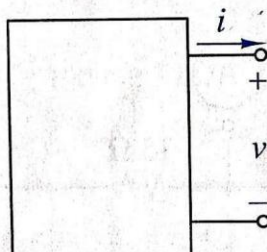


FIGURE P3-42

3-43 The i - v characteristic of the active circuit in Figure P3-42 is $5v + 500i = 100$. Find the output voltage when a $1\text{-k}\Omega$ resistive load is connected.

3-44 You have successfully completed Circuits I, and as part of an undergraduate work-study program your former professor has asked you to help him with Circuits I and